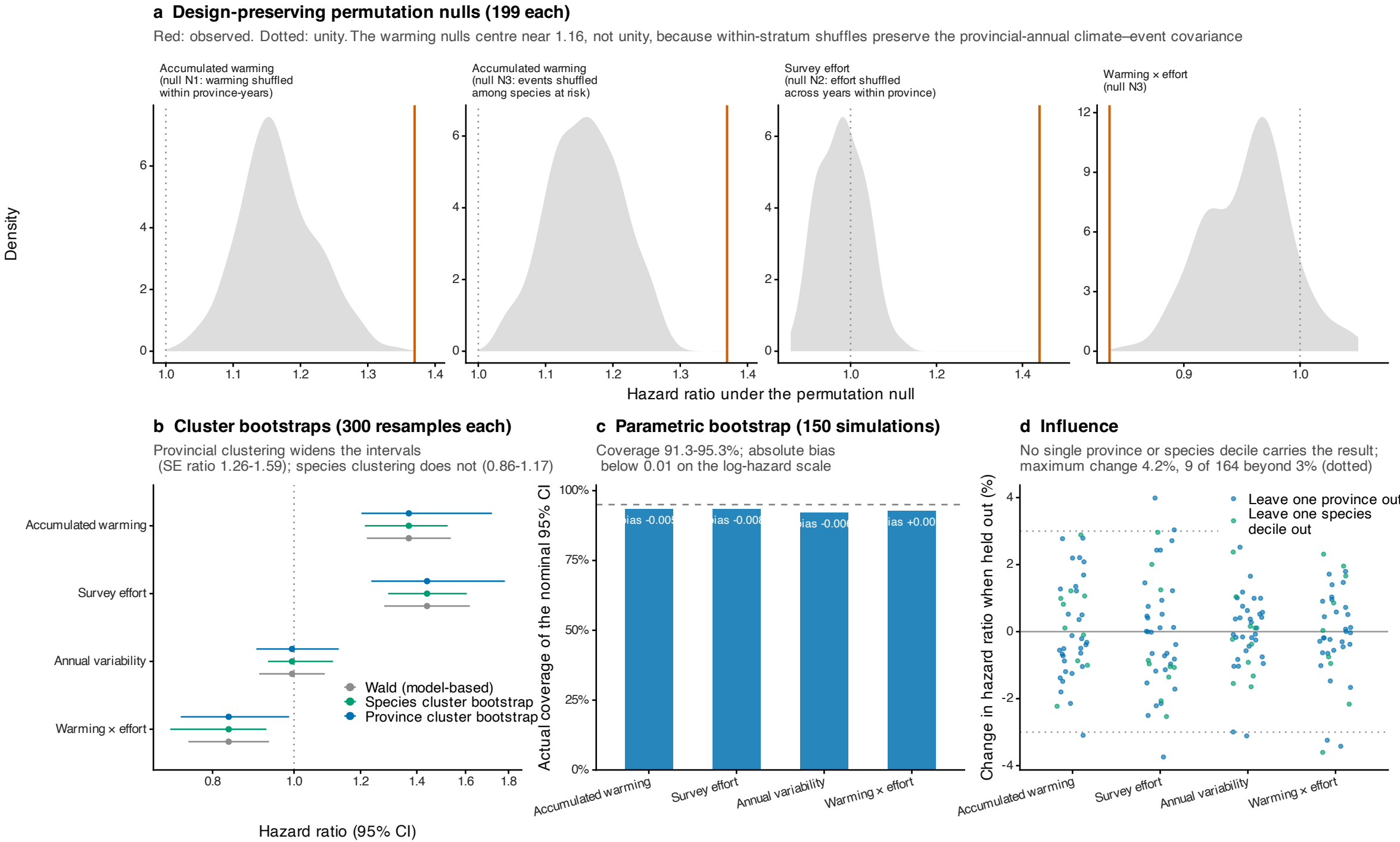
a Design-preserving permutation nulls (199 each)
Red: observed. Dotted: unity. The warming nulls centre near 1.16, not unity, because within-stratum shuffles preserve the provincial-annual climate–event covariance
Accumulated warming (null N1: warming shuffled within province-years)
Accumulated warming (null N3: events shuffled among species at risk)
Survey effort (null N2: effort shuffled across years within province)
Warming × effort (null N3)
Density
Hazard ratio under the permutation null
b Cluster bootstraps (300 resamples each)
Provincial clustering widens the intervals (SE ratio 1.26-1.59); species clustering does not (0.86-1.17)
Accumulated warming
Survey effort
Annual variability
Warming × effort
Wald (model-based)
Species cluster bootstrap
Province cluster bootstrap
Hazard ratio (95% CI)
c Parametric bootstrap (150 simulations)
Coverage 91.3-95.3%; absolute bias below 0.01 on the log-hazard scale
Actual coverage of the nominal 95% CI
bias -0.005
bias -0.008
bias -0.006
bias +0.00
d Influence
No single province or species decile carries the result; maximum change 4.2%, 9 of 164 beyond 3% (dotted)
Change in hazard ratio when held out (%)
Leave one province out
Leave one species decile out